Sample Papers

	FINALTERM EXAMINATION SPRING 2007 CS402 - THEORY OF AUTOMATA (Session - 5)	Marks: 55 Time: 120min

StudentID/LoginID: ______________________

Student Name: ______________________

Center Name/Code: ______________________

Exam Date: Saturday, July 07, 2007

1. **This examination is closed book, closed notes, closed neighbors.**
2. **Answer all questions.**
   - a. **There is no choice.**
   - b. **You will have to answer correctly all questions in this examination to get the maximum possible marks.**
3. **Do not ask any questions about the contents of this examination from anyone.**
   - a. **If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**
   - b. **If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.**
4. **Use of cell phone during the examination is strictly prohibited, otherwise strict disciplinary action will be taken as per university rules**
5. **You are allowed to use only MS Word for drawing Diagrams and Symbols.**

www.ilmkidunya.com

**WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.

For Teacher's use only

Question	1	2	3	4	5	6	7	8	9	10	Total
Marks											
Question	11	12	13	14	15						
Marks											
Question											
Marks											

Question No: 1 (Marks: 1) - Please choose one

PDA is only used to represent a regular language.

- ▶ True
- ▶ False

Question No: 2 (Marks: 1) - Please choose one

If L is a regular language then L^C is also a regular language.

- ▶ True
- ▶ False

Question No: 3 (Marks: 1) - Please choose one

A production of the form non-terminal → string of two non-terminal is called a live Production.

- ▶ True
- ▶ False

Question No: 4 (Marks: 1) - Please choose one

we can find a CFG corresponding to a DFA.

- ▶ True
- ▶ False

Question No: 5 (Marks: 1) - Please choose one

START, READ, HERE and ACCEPTS are conversions of the machine

- ▶ True
- ▶ False

Question No: 6 (Marks: 1) - Please choose one

A CFG is said to be ambiguous if there exists at least one word of its language that can be generated by different production trees

- ▶ True
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

Syntax tree or Generation tree or Derivation tree are same tree

- ▶ True
- ▶ False

Question No: 8 (Marks: 1) - Please choose one

The symbols that cannot be replaced by anything are called terminals

- ▶ True
- ▶ False

www.ilmkidunya.com

Question No: 9 (Marks: 1) - Please choose one

The production of the form non-terminal → one non-terminal is called unit production

- ▶ True
- ▶ False

Question No: 10 (Marks: 1) - Please choose one

DFA and PDA are equal in power.

- ▶ True
- ▶ False

Question No: 11 (Marks: 10)

a) Define Describe the following terms:
[Note: maximum in 20 to 30 words for each]

I. CNF
II. Regular Grammar

b) Convert the following CFG into CNF

$S \rightarrow CDCD$
$C \rightarrow 0|\Lambda$
$D \rightarrow 1|\Lambda$

Question No: 12 (Marks: 10)

a) Construct RE's for following languages over $\Sigma = \{0, 1\}$ (6)

i. All words in which "1" never follows "0". ("0" never appears before "1")

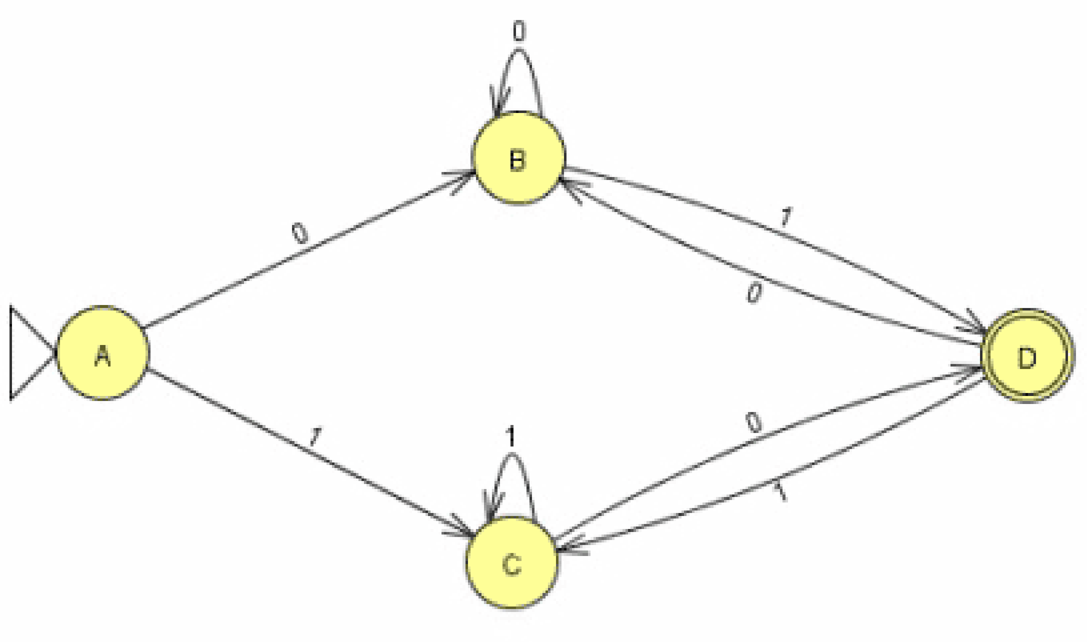
ii. All words which begin and end with different letters.

b) How many minimum states can be there in an NFA of language having all word with "101" at the end? (4)

HINT: Build NFA for language for yourself but write only number of minimum states. No need to build NFA in software.

Question No: 13 (Marks: 10)

a) Is this an FA or NFA? (2)



b) Determine the CFG corresponding to the above FA or NFA (8)

Question No: 14 (Marks: 10)

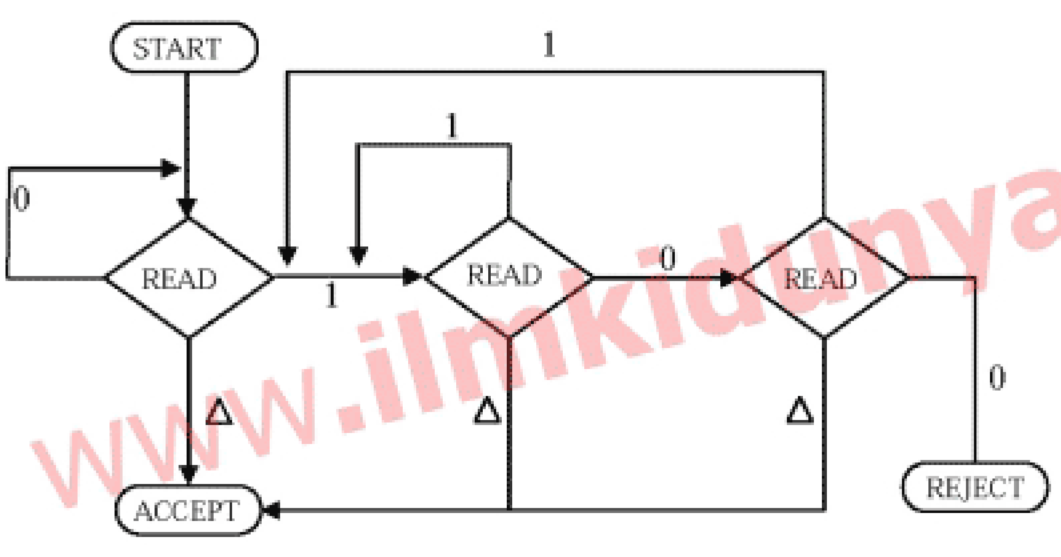
a) Given CFG (Context Free Grammar): (6)

$S \rightarrow bS \mid aX \mid \Lambda$
$X \rightarrow aX \mid bY \mid \Lambda$
$Y \rightarrow aX \mid \Lambda$

Derive following strings from above CFG. Show all steps. If string cannot be derived then describe it.

i. baabab
ii. ababaab

b) Describe language of following PDA (Push Down Automata): (4)
[Note: Don't write more than two to three lines for each. Only write to the point.]



www.ilmkidunya.com

Question No: 15 (Marks: 5)

Given a CFG below, [5]

$S \rightarrow bS \mid aM$
$M \rightarrow bM \mid aF$
$F \rightarrow bF \mid aM \mid$ ^

What is language accepted by above CFG?